Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: **Bachelor of Science in Sanitary Engineering**

Primary Job Roles students are prepared for in the program	**Sanitary Engineer, Environmental Planner/Engineer, Construction Manager/Engineer, Design Engineer, Project Manager/Engineer, Consultant**
Secondary Job Roles students are prepared for in the program	**Pollution Control Officer, Public Health Engineer, Safety Engineer, Industrial Hygiene Engineer, Hospital Administrator, Educator/Academician, Public Service Administrator, Researcher**

Primary and Secondary Job Roles provide a mental anchor to the identification of program outcomes you will be having for your program. This will allow you to check for completeness in your program outcomes.

Current CMO Guiding Program: CMO 98, Series of 2017

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of policies, standards and guidelines in the establishment of an outcomes-based education (OBE) system in higher education institutions offering engineering programs as advocated under CMO 37 s. 2012 and policy-standard to enhance quality assurance (QA) in Philippine higher education through an outcomes-based and typology-based QA as advocated under CMO 46 s. 2012, and by virtue of Commission en banc Resolution No. 788-2017 dated October 24, 2017 the following policies and standards are hereby adopted and promulgated by the Commission.

This CMO integrates the establishment of an outcomes-based Based on the Guidelines for the Implementation of CMO No. 46 series of 2012 and CMO 37 s. 2012, this PSG implements the shift to outcomes-based education (OBE) leading to competency-based standards. It specifies the "core competencies" expected of Bachelor of Science in Sanitary Engineering (BSSE) graduates "regardless of the type of Higher Education Institutions (HEI) they graduate from." However, in recognition of outcomes-based education and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description (CMO 98 Series of 2017):

Degree Name
The degree program described herein shall be called Bachelor of Science in Sanitary Engineering (BSSE).

Nature of the Field of Study
The BSSE program recognizes the importance of sanitary engineering in the protection and preservation of the environment as well as the promotion of health and well-being of the general public. The groundwork of student preparation is based on acknowledging the fundamental competencies required of the graduates to assume the specific roles of professionals in nation building in the future.

Graduates of this program are expected to acquire entry-level competency at par with those of the neighboring countries in the region by utilizing curriculum structure and methods that would cater to national and international settings. The curriculum shall lead the students to increase knowledge and skills not only on basic concepts and principles but also exposure to current trends of practice of the profession in the design, construction, management and evaluation of Sanitary Engineering programs and projects. The program shall likewise emphasize techniques to nurture and sustain competency through individual and institution-based continuing education activities, transfer of knowledge and skills, entrepreneurial and managerial attitude and skills.

Social values and professional ethics are integrated with the science and technological courses to enable the students to internalize and actualize social accountabilities and humane exercise of practice of the profession. The curriculum will keep abreast the graduates with the economic, technological and scientific modernization and globalization initiatives or updates in the profession to optimize the impact of this program to human society.

VISION
A PREMIER UNIVERSITY IN THE ASEAN REGION

MISSION
USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.

CORE VALUES
USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

Program Educational Objectives		Mission-Vision and Core Value Alignment						Reference Source or CMO Provision Compliance
		Vision	Mission	Core Value				
				U	S	E	P	
PEO1	Demonstrate expertise in the field of Sanitary Engineering	√	√		√	√	√	USEP Brochure, CMO 98
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√		√	USEP Brochure, CMO 98
PEO3	Demonstrate gender sensitive leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√		√	√	USEP Brochure, CMO 98
PEO4	Exhibit globally competitive skills	√	√		√	√	√	USEP Brochure, CMO 98

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting statements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Institutional Graduate Attributes:

IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

Institutional Graduate Attribute	Program Outcomes		Performance Indicators	
IGA1	**a**	Apply knowledge of mathematics and sciences to solve engineering problems;	1	Apply concepts in mathematics to solve engineering problems
IGA4			2	Apply principles in chemical sciences to solve engineering problems.
IGA6			3	Apply principles in physical sciences to solve engineering problems
IGA1, IGA4	**b**	Design and conduct experiments, as well as to analyze and interpret data	1	Perform laboratory experiments as prescribed by standard procedures.
IGA5, IGA6			2	Design an experiment to achieve a specific objective
IGA1, IGA2			1	Develop an engineering solution that have various constraints as factors

IGA3, IGA4 IGA5, IGA6	**c**	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	2	Design a system or process based on various constraints
IGA2	**d**	Function on multidisciplinary teams	1	Shows tact in dealing with people he/she comes in contact with.
IGA3			2	Capable of learning from and listening to co-members
IGA6			3	Shows respect and courtesy in dealing with peers and superiors.
IGA1	**e**	Identify, formulate, and solve engineering problems	1	Interpret statement problems into mathematical expressions.
IGA4			2	Apply concepts, theories, and principles in solving engineering problem(s)
IGA6			3	Solve specific engineering problem related to discipline.
IGA4	**f**	Understand professional, social and ethical responsibility	1	Internalizes the professional code of ethics for specific discipline.
IGA5			2	Applies the principles of professional ethical standard and social responsibility to actual practice
IGA6				
IGA4	**g**	Communicate effectively	1	Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5			2	Articulate the convention and style in written communication that conveys complete thought and information.
IGA6			3	Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3	**h**	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1	Relate engineering solutions to global, economic, environmental, and societal context.
IGA4, IGA6			2	Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2	**i**	Recognize the need for, and an ability to engage in life-long learning	1	Demonstrate ability to learn independently.
IGA3, IGA4			2	Shows appreciation and understanding of the engineering profession.
IGA5, IGA6			3	Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3	**j**	Apply knowledge of contemporary issues	1	Identify the socio-economic, political, and environmental issues related to the practice of engineering
IGA4, IGA5			2	Explain the possible impact of technology to local, national, and international issues
IGA6			3	Shows leadership skills in managing a multidisciplinary group.
IGA1	**k**	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
IGA4			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
IGA6			3	Use techniques, skills, and modern engineering tools necessary for engineering practice
IGA1, IGA2	**l**	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
IGA3, IGA4			2	Shows leadership skills in managing a multidisciplinary group.
IGA5, IGA6			3	Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

				Program Outcomes											
Year Level	**Semester**	**Units**	**Courses in the Curriculum**	**a**	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**
FIRST YEAR	**FIRST**	3	Purposive Communication												
		3	Mathematics in the Modern World												
		3	The Life and Works of Rizal												
		3	Engineering Calculus 1	I				I							
		4	Chemistry for Engineers	I	I			I							
		1	Engineering Drawing and Plans				I			I					
		2	Sanitary Engineering Orientation						I		I				
		2	Movement Enhancement												
		3	National Service Training Program 1												
	SECOND	3	Understanding the Self												
		3	Ethics												
		3	Science, Technology and Society												
		3	Engineering Calculus 2	E				E							
		4	Physics for Engineers	E	I		I	E							
		1	Computer Aided Drafting						I	I		I		I	
		3	Environmental and Sanitary Chemistry	I				I			I				
		1	Environmental and Sanitary Chemistry, Lab		I						I				
		2	Fitness Exercise												
		3	National Service Training Program 2												
SECOND YEAR	**FIRST**	3	Readings in Philippine History												
		3	Contemporary World												
		2	Engineering Geology	E				E			E				
		3	Differential Equations	E				E							
		3	Engineering Data Analysis	I				I			I				
		3	Computer Fundamentals and Programming		I							I		I	
		3	Engineering Economics	E					I		I		I		I
		5	Mechanics of Rigid Bodies	E				E							
		2	Physical Activities Towards Health & Fitness I												

				Program Outcomes											
Year Level	Semester	Units	Courses in the Curriculum	a	b	c	d	e	f	g	h	i	j	k	l
SECOND YEAR	SECOND	3	Arts Appreciation												
		4	Mechanics of Deformable Bodies	E				E							
		3	Methods of Research						E	E	E		E		E
		3	Environmental Science and Engineering						E		E		E		E
		3	Fundamentals of Surveying		E		E	I		E				E	
		2	Fundamentals of Surveying, Field		E		E	I		E				E	
		3	Basic Mechanical Engineering	E				E							
		3	Basic Electrical Engineering	E				E							
		2	Physical Activities Towards Health & Fitness II												
THIRD YEAR	FIRST	3	Numerical Solutions to SE Problems	E				E		E		E		E	
		3	Geotechnical Engineering, Lecture	E	E	E	E	E		E			E	E	
		1	Geotechnical Engineering, Lab	E	E	E	E	E		E			E	E	
		4	Structural Theory	E	E			E						E	
		4	Hydraulics, lecture	E	E		E	E		E					
		1	Hydraulics, lab	E	E		E	E		E					
		3	Hydrology	E				E		E				E	
		3	Microbiology and Parasitology for Environmental Engineers	E					E		E		E		
		3	Solid and Hazardous Waste Engineering			E		E			E	E			
	SECOND	2	Construction Materials and Testing, lecture	E	E		E			E			E	E	
		1	Construction Materials and Testing, lab	E	E		E			E			E	E	
		4	Principle of Reinforced / Pre-stressed Concrete	E	E									E	
		3	Sanitary Science, Fire Protection and Plumbing as Applied to Building	E	E	E						E		E	E
		3	Water Supply Planning and Development	E		E			E	E				E	
		3	Sewerage and Urban Drainage	E	E	E						E		E	E
		3	Basic Occupational Health and Safety					E	E	E			E		
		2	Public Health Engineering				E		E	E	E	E	E		
		3	Environmental Planning, Laws and Impact Assessment						E			E	E		E
		3	General Education Elective 1												
	OFF-SEMESTER	3	Engineering Management				E		E	E	E		E		E
		3	Technopreneurship				E		E	E					E

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes											
				a	b	c	d	e	f	g	h	i	j	k	l
FOURTH YEAR	FIRST	3	General Education Elective 2												
		3	General Education Elective 3												
		3	SE Project 1		D	D		D		D	D	D	D	D	D
		3	Water Purification Process Design	D		D			D				D	D	
		3	Sewerage and Industrial Wastewater Treatment	D		D			D				D	D	
		3	Sanitary Engineering Laws, Contract and Ethics						D	D	D	D			
		1	Environmental Engineering laboratory		E	E					E			E	
		2	Planning and Design in Environmental and Sanitary Engineering		D	D		D		D		D			
		4	Project Management				D		D	D	D		D		D
FOURTH YEAR	SECOND	2	Correlation Course	E				E				E			E
		1	FREE ELECTIVE	E				E				E			
		2	Specialized On-the-Job Training (320 hrs)				D		D	D		D	D	D	

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

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